

1 Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the latent space learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the latent representations produced by the neural network and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representations (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the concept structure of the latent space. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

The remainder of this chapter is organized as follows. The first section introduces the theoretical assumptions underlying the proposed CAR method, referred to as the representation convergence hypothesis. The second section describes the CAR method itself, with a detailed account of the extraction of latent representations using surrogate policies, along with the automatic detection of concept structures via clustering. The third section then presents the proposed evaluation metric.

1.1 Representation Convergence Hypothesis

The Clustering Agent Representations (CAR) method relies on a assumption, referred to as the Representation Convergence Hypothesis. This hypothesis follows the two assumptions established previously throughout this manuscript.

The first assumption rejects the notion that deep neural networks constitute irreducible black boxes ???. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be highly complex, they remain explainable through the analysis of the internal mechanisms.

The second assumption states that neural networks achieve generalization by learning concepts ?. During optimization, the network progressively organizes its latent space into representations that capture the information required to solve the task.

Following these hypotheses, it becomes possible to explain the Neural Network (NN)’s process by understanding the organization of its latent space. This family of approaches is commonly referred to as mechanistic interpretability.

In mechanistic interpretability, the consistency with which a concept is recovered across different models or independent extractions is referred to as its universality [Templeton et al., 2024]. A concept is said to be universal if it repeatedly emerges despite variations in model initialization, optimization, or training conditions.

The Representation Convergence Hypothesis provides a theoretical explanation for this phenomenon. We hypothesize that, for a given task and a given neural network architecture, there exists a privileged conceptual organization. This organization naturally emerges because it constitutes an efficient representation of the information required to solve the task. Under this assumption, the universality of a concept is not merely an empirical observation but rather the expected consequence of a common underlying conceptual structure shared by independently learned representations.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a more abstract level of representation. We assume that, although the numerical implementation of the networks may differ, the conceptual organization induced by their Latent Representation (LR) converges toward an equivalent conceptual structure. Consequently, the objective is not to compare neural parameters or latent activations directly, but compare the underlying conceptual structure encoded within the latent space.

The remainder of this chapter builds upon this hypothesis. This principle forms the theoretical foundation of the *Clustering Universality* metric introduced later in Subsection 1.3.3.

1.2 Method

As discussed in Chapter ??, several hypotheses have been proposed regarding the organization of latent representations in deep neural networks. Following the work of [Zahavy et al., 2017], we therefore analyze the latent space through clustering techniques, placing our approach within the broader field of Deep Clustering (Section ??).

This design choice is motivated by our rejection of the superposition hypothesis and, more generally, of any view that assumes a linearly organized concept structure. We consider this premise unnecessarily restrictive.

The CAR methodology is composed of two main stages. The first stage

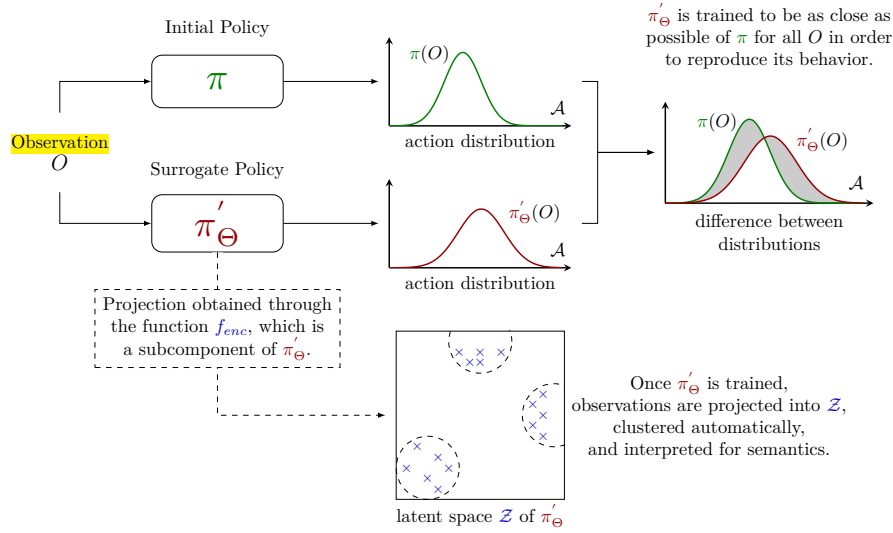


Figure 1.1: Illustration of the Clustering Agent Representation method.

consists of generating multiple independent neural networks that implement the same decision policy, referred to as surrogate policies. The second stage clusters their latent representations in order to identify the underlying conceptual organization.

This section is organized as follows. In the first subsection, we justify the use of multiple surrogate policies in order to study the stability of representations. In the second subsection, we detail how these surrogate policies are trained so as to best match the original policy.

1.2.1 Multiple surrogate policies

The CAR method was developed with the aim of verifying the representation convergence hypothesis presented in Section 1.1. To test this hypothesis, several neural networks must be trained independently and their latent representations compared. If similar conceptual structures consistently emerge, this provides empirical evidence supporting the Representation Convergence Hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. Reinforcement learning, introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. Making it impossible to determine whether differences observed in the latent space originate from the learning dynamics or from genuine differences between the learned policies.

To overcome this limitation, we do not directly compare independently trained reinforcement learning agents. Instead, we first consider a initial policy that has already been trained. This policy is then imitated using several independently initialized surrogate neural networks trained through supervised behavioral cloning. Since all surrogate models approximate the same target function, they

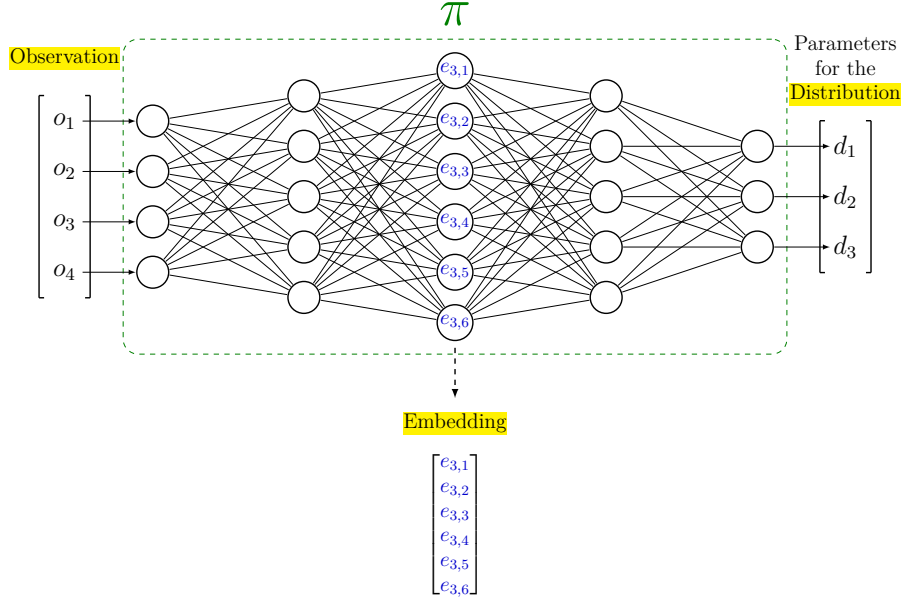


Figure 1.2: Architecture of the surrogate policy π : the encoder maps the observation O to the embedding Z , from which the decoder reconstructs the action distribution parameters.

are expected to converge toward equivalent decision policies while preserving independent internal parameterizations. It becomes possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection ?? . The surrogate policies remain deep neural networks. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence. The explanations are therefore still extracted directly from the internal representations of deep neural networks, making CAR a fully intrinsic and mechanistic explainability approach.

1.2.2 Training surrogate policies

Throughout the remainder of this manuscript, the initial policy we aim to explain is denoted as π (represented in green in Figure 1.1). This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [?], alpha-beta algorithm [?], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π .

The replicated policies are referred to as surrogate policies, denoted by π' (shown in red in Figure 1.1). For any observation O , the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a NN parameterized by Θ . This

network is composed of two principal components: an encoder and a decoder. The encoder, represented by the function f_{enc} , maps observations from $\mathcal{O}$ into a Latent Space (LS) $\mathcal{Z}$ (shown in blue in Figure 1.1). The decoder then reconstructs actions from this latent representation. Figure 1.2 details this architecture, showing how the encoder layers progressively shape the observation into the embedding before the decoder reconstructs the action distribution parameters from it.

To train π' to accurately replicate the action distribution, we define a global loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section ???. This loss, defined in Equation (1.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function p , which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π .

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}(p(\pi(O)), p(\pi'_{\Theta}(O))) \quad (1.1)$$

We denote by $\mathcal{E}$ the set of embeddings obtained by applying f_{enc} to a subset of observations $\mathbb{O} \subseteq \mathcal{O}$.

On the set $\mathcal{E}$, we use a clustering function f_{clust} to partition the point cloud of embedded observations into distinct clusters. Each observation is assigned a cluster label based on its proximity in the LS, resulting in the clustering $\mathcal{C}$.

To demonstrate this clustering faithfully reflects the agent’s internal decision-making process, we introduce a set of evaluation metrics, which will be detailed in the following section.

1.3 Evaluation Metrics

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- **Behavior Reconstruction** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- **Clustering Universality** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- **Abstraction Score** : it quantifies the extent to which LS clusters capture behavioral abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and describes how they are calculated.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \dots, \pi'_n$, trained as described in Section 1.2. Each surrogate policy is associated with a projection function

173 $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embed-
 174 dings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set,
 175 producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

176 1.3.1 Behavior Reconstruction

177 We argue that explaining a function b that closely approximates another func-
 178 tion a is essentially the same as explaining a . In other words, if b accurately
 179 captures the key decision patterns of a , understanding b provides meaningful
 180 insights into how a works [Guidotti et al., 2018]. In our setting, this means that
 181 generating explanations for π' that faithfully reproduce π amounts to explaining
 182 π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior*
 183 *Reconstruction* metric.

184 This metric is computed by comparing the cumulative rewards obtained by
 185 the original policy π and the surrogate policies π' over m evaluation episodes.
 186 The behavior reconstruction metric is then defined as:

$$\text{Behavior Reconstruction} = \frac{1}{n \sum_{i=1}^m g_{\pi}^i} \sum_{j=1}^n \sum_{k=1}^m g_{\pi'_j}^k \quad (1.2)$$

187 In imitation learning, it is unlikely to expect the surrogate policy π' to
 188 outperform the initial policy π it aims to approximate through supervised
 189 learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$.
 190 Values approaching 1 indicate minimal loss in reproducing π , while values close
 191 to 0 indicate poor reproduction.

192 We deliberately adopt a reward-based definition rather than alternatives based
 193 on actions or action distributions (e.g., training loss, accuracy, KL divergence
 194 [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate
 195 policies are trained in a supervised manner, they cannot develop strategies
 196 beyond those demonstrated by the initial policy. This means that if the rewards
 197 achieved are comparable, the surrogate policy is effectively copying the original
 198 policy’s behavior.

199 1.3.2 Adjusted Rand Index

200 The previous section established that the surrogate policies π' faithfully reproduce
 201 the behavior of the original policy π through the *Behavior Reconstruction*
 202 procedure. Having controlled for behavioral differences, we can now focus on
 203 the internal organization of their latent representations.

204 As discussed in Chapter ??, CAR assumes that latent concepts are organized
 205 as clusters within the representation space. Under the Representation Conver-
 206 gence Hypothesis, independently trained surrogate policies implementing the
 207 same decision function are expected to develop similar conceptual organizations.
 208 Importantly, this similarity is not expected at the level of the latent coordinates
 209 themselves, since independently trained neural networks may learn different
 210 embeddings of the same concepts. Instead, the similarity should be observed
 211 at the level of the partition induced by the clustering process: observations
 212 corresponding to the same underlying concept should consistently be assigned
 213 to the same cluster across independent models.

Comparing two clusterings therefore becomes a central operation in the proposed methodology. This comparison should be independent of the geometric arrangement of the latent space and depend only on the grouping of observations into concepts. In the clustering literature, one of the most widely used measures for comparing two partitions is the Adjusted Rand Index (ARI). The ARI quantifies the agreement between two clusterings solely based on their assignments, irrespective of the spatial position of the clusters.

Both metrics introduced in this chapter, namely *Clustering Universality* and *Abstraction Score*, are built upon the Adjusted Rand Index. Since the behavior of the ARI is often non-intuitive, we first review its definition and discuss its main properties before introducing our proposed metrics.

The Adjusted Rand Index (ARI) [Hubert and Arabie, 1985] measures the similarity between two partitions of the same dataset. Unlike geometric clustering metrics, which compare the spatial arrangement of clusters, the ARI evaluates only whether pairs of observations are assigned consistently across two clusterings. As such, it is particularly well suited to our objective, since CAR aims to compare conceptual partitions rather than the geometry of latent representations.

The ARI is derived from the Rand Index (RI) [Rand, 1971], which measures the agreement between two partitions by considering every possible pair of observations. For each pair, four situations are possible: the observations may belong to the same cluster in both partitions, to different clusters in both partitions, or they may disagree by being grouped together in only one of the two partitions. The Rand Index simply measures the proportion of pairs for which the two partitions agree. Formally, it is defined as:

$$\text{RI} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}.$$

Although the Rand Index is intuitive, it suffers from an important limitation. In most clustering problems, the vast majority of pairs of observations belong to different clusters. Consequently, even two completely unrelated partitions will agree on a large number of pairs simply because both assign them to different clusters. As the number of observations increases, this effect becomes more pronounced, causing the Rand Index to report relatively high similarity scores even when the clusterings are essentially independent.

The Adjusted Rand Index (ARI) addresses this issue by accounting for the agreement that would be expected purely by chance. Rather than measuring the raw proportion of agreeing pairs, it compares the observed agreement to the agreement obtained by random partitions having the same cluster size distribution. The score is then normalized by the maximum agreement achievable beyond this random baseline. As a result, an ARI of (0) corresponds to the level of agreement expected from random clusterings, while an ARI of (1) indicates perfect agreement between the two partitions:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

The previous equations provide an intuitive understanding of the Adjusted Rand Index. In practice, the ARI is computed directly from the cluster assignments of the two partitions. Let clustering A be composed of partitions $\{A_i\}$ and clustering B of partitions $\{B_j\}$. Let $a_i = |A_i|$ denote the number of samples

in partition i of clustering A , and $b_j = |B_j|$ the number of samples in partition j of clustering B . Furthermore, let $n_{ij} = |A_i \cap B_j|$ be the number of samples shared by partition i of clustering A and partition j of clustering B , and let n be the total number of samples. With these definitions, the ARI can be computed as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}$$

The ARI is a widely used metric, but it presents several non-intuitive properties that complicate its interpretation:

- **Non-Linearity:** The ARI does not scale linearly with perceived clustering quality. An increase from 0.2 to 0.4 does not represent the same improvement as an increase from 0.7 to 0.9. Its scale is compressed in the middle range, making differences near 0 or 1 more significant than they appear.
- **Dependence on Number of Clusters:** The range of *achievable* ARI values is constrained by the number of clusters in the two partitions being compared. For a fixed number of true classes, an algorithm that proposes a very different number of clusters is mathematically capped from achieving a high ARI, regardless of how semantically meaningful the subdivision might be.
- **Asymmetry in Penalties:** The ARI penalizes false splits (splitting a true cluster into multiple parts) and false merges (merging multiple true clusters into one) asymmetrically. The exact penalty structure is complex and not easily intuited from the formula, making it difficult to diagnose what specific type of error is driving a low score.
- **Lack of Intuitive Baseline:** While an ARI of 0 corresponds to the expected similarity of random label assignments, this “random” baseline is itself a function of the number and sizes of clusters. Therefore, the meaning of “0” is context-dependent.

The ARI is best used for comparative assessment (e.g., Algorithm A vs. Algorithm B on the same data) rather than as an absolute measure of quality. A score of 0.5 should be interpreted as “substantially better than random and often pragmatically useful,” not as “halfway to perfect” [Amezquita et al., 2020]. Care should be taken when comparing ARI values across different datasets or problems, as the feasible upper bound and the meaning of a given score are highly context-dependent. Now that the ARI has been introduced, we can present the two other metrics based on it.

1.3.3 Clustering Universality

Our methodology involves analyzing the representations of an agent. Following the representation convergence hypothesis introduced in Section 1.1, for these representations to serve as a valid object of study in explainable artificial intelligence, they must demonstrate consistency across independent training runs. Indeed, if each surrogate π' produces substantially different representations, it

is meaningless to search for recurring structures to interpret. Conversely, if independently trained surrogates yield consistent representations, the resulting structures are more likely to reflect intrinsic properties of the original policy rather than artifacts of training.

In our study, agent representations take the form of clusters in the latent space. To assess the stability of these representations, we measure the similarity of clusters produced by different surrogate policies π' using the ARI [Hubert and Arabie, 1985]. For this reason, we refer to our metric as *Clustering Universality*, which evaluates the similarity between clusterings as follows:

$$\text{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^n \sum_{j=0}^{i-1} \text{ARI}(\mathcal{C}_i, \mathcal{C}_j) \quad (1.3)$$

This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate policies produce highly similar partitions, suggesting that the clusters reflect intrinsic properties of the policy. Conversely, a value near 0 implies that the similarity between partitions is no better than that of two randomly generated clusterings, meaning that the clustering fails to capture any meaningful structure in the policy. A negative value, on the other hand, signals that the agreement between partitions is worse than what would be expected by chance, indicating a systematic disagreement between the clusterings.

For further details on the properties of the ARI, we refer the reader to [Hubert and Arabie, 1985, Warrens and van der Hoef, 2022] and to Section ?? . Nevertheless, it is generally accepted that an ARI score above 0.5 indicates a reasonably strong agreement between two clusterings [Amezquita et al., 2020].

1.3.4 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

- (i) the LS merely reproduces the structure of the observation space or action distribution, offering no abstraction, as defined in Definition ??;
- (ii) the LS captures similarities in decision-making processes, yielding behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with reference structures derived from the raw observation and action data. We consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are performed using the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering procedure itself are minimized.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using the ARI:

$$\begin{aligned} A_i^{obs} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs}) \\ A_i^{act} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act}) \end{aligned}$$

335 The *Abstraction Score* is subsequently calculated as one minus the average of
 336 the maximum absolute similarity values across these two reference baselines:

$$Abstraction\ Score = 1 - \frac{1}{n} \sum_{i=1}^n \max(|A_i^{obs}|, |A_i^{act}|) \quad (1.4)$$

337 The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the
 338 latent clusters are strongly aligned with either the observation or the action
 339 space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that
 340 the latent clusters are independent of both the observation and action spaces,
 341 capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

342 The following chapter presents the results of applying these metrics to specific
 343 examples.

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